A Tale of Five Decoders

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November 6, 2025

Introduction

Today:

- Recap and where we stood before the summer.
- Simulations results.
- Where we heading.

Map

Coming with $O(1)$-depth
gates for 'good' codes is hard.

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$\downarrow$

Is the (d, k)-Toric a MEM?

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Dennis at el? [Den+02]

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Swift-Rule? [KP19]

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Is the (d, k)-Toric a MEM?

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Dennis at el? [Den+02]

$\downarrow$

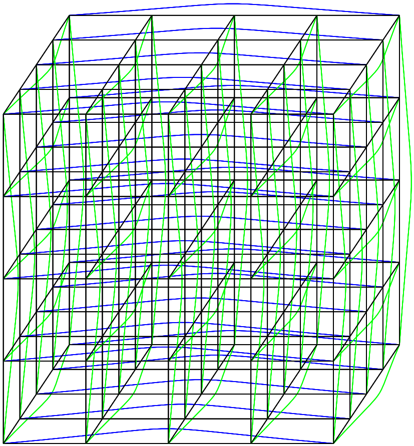
Swift-Rule? [KP19]

$\downarrow$

Simulations

The Code

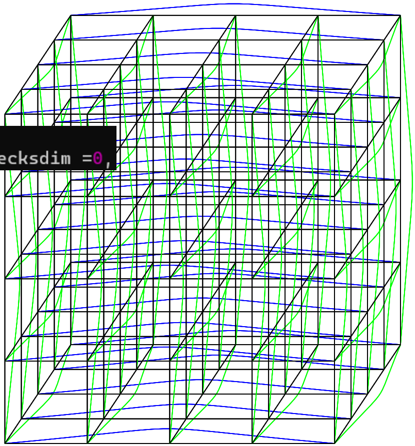
The simulations are over the 3D-Toric
where qubits set on faces
and checks on vertices.



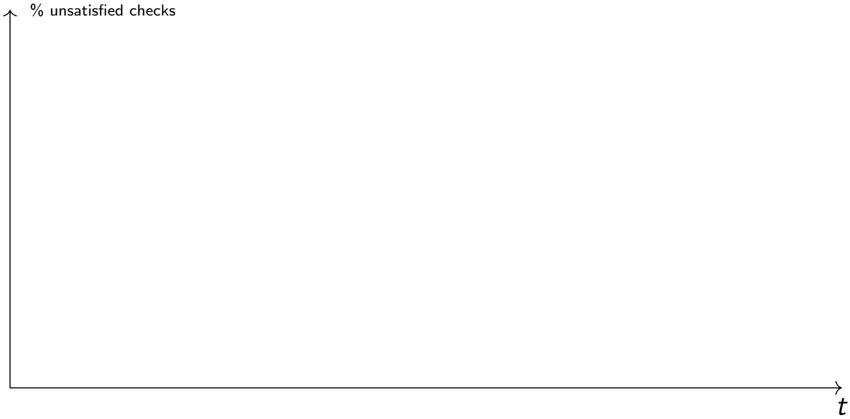
The Code

The simulations are over the 3D-Toric
where qubits set on faces
and checks on vertices.

```
class KDgrid:
    def __init__(self, n, k, celldim=2, checksdim =0,
```

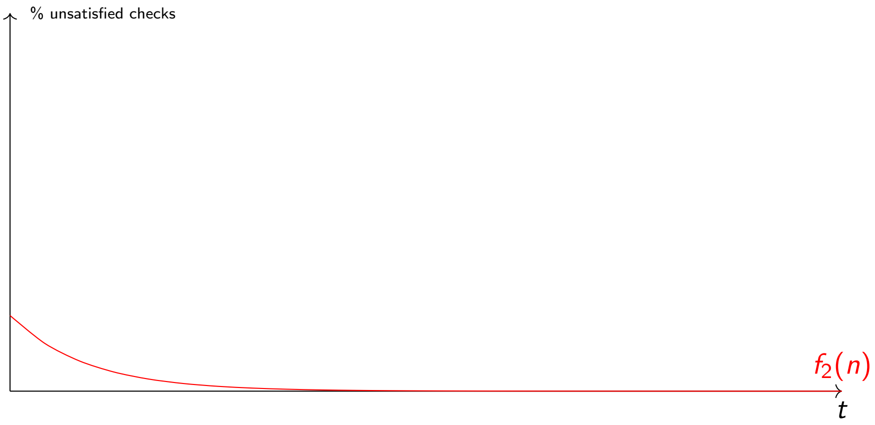


How to Read



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

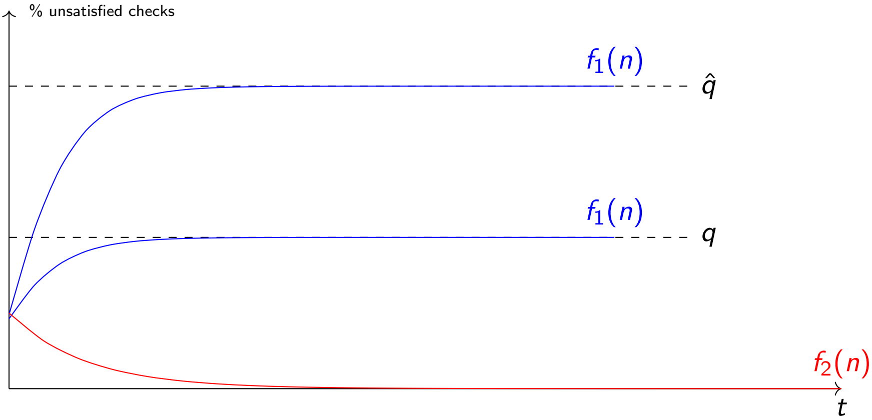
Ideal Result.



colors

In red, No accumulation of errors.

Ideal Result.

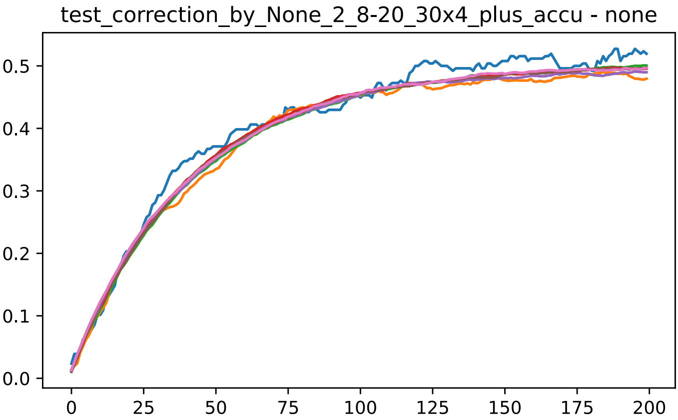


colors

In red, No accumulation of errors.

In blue decoding in the presence of noise.

Without Decoding



side-length :[8, 16, 30, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True

Majority

Alg.

Each bit looks at its neighbour checks, if the majority of them is unsatisfied, then it flip itself.

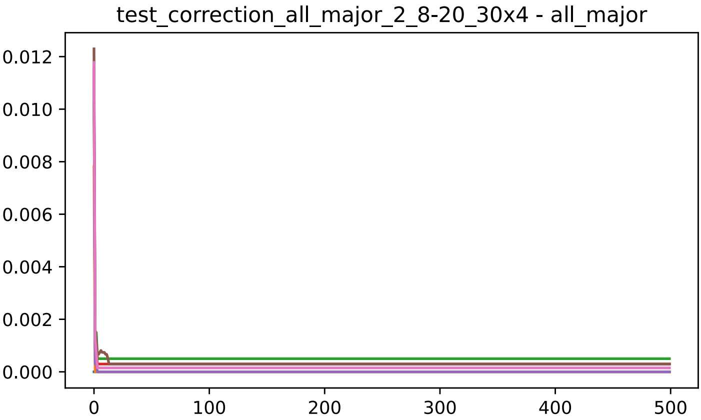
Justification.

Was suggested in Dennis.

Bottom line.

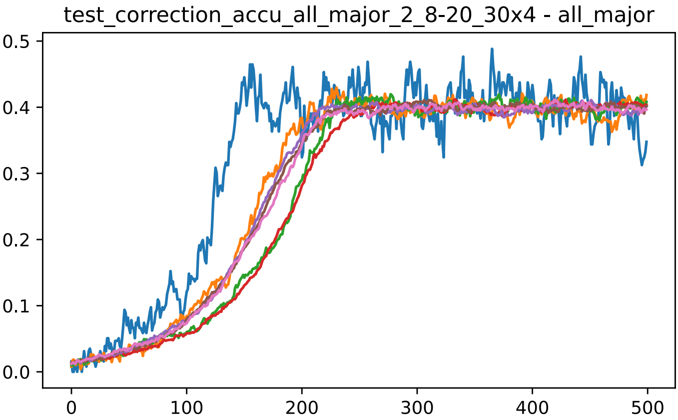
- Collisions.
- Gives better result when requiring $(\frac{1}{2} + \mu)$-majority.

Majority



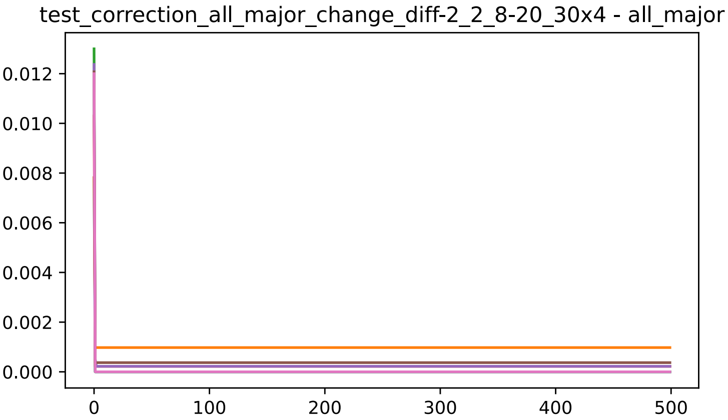
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Majority



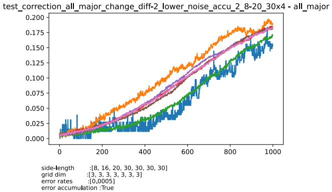
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True

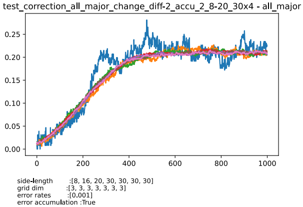
μ-Majority

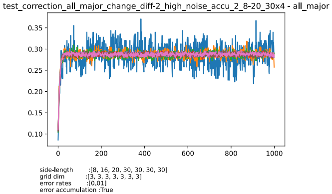


side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

μ-Majority


test_correction_all_major_change_diff-2_lower_noise_accu_2_8-20_30x4 - all_major
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
0
200
400
600
800
1000
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.0005]
error accumulation :True


test_correction_all_major_change_diff-2_accu_2_8-20_30x4 - all_major
0.25
0.20
0.15
0.10
0.05
0.00
0
200
400
600
800
1000
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True


test_correction_all_major_change_diff-2_high_noise_accu_2_8-20_30x4 - all_major
0.35
0.30
0.25
0.20
0.15
0.10
0
200
400
600
800
1000
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.01]
error accumulation :True

Colors

Alg.

At the initialization, We divide the bits into subsets (colors). Bits of the same color don't share common check. Then, at the correction round ith only bits associated with the color i mod **COLORS** apply a correction.

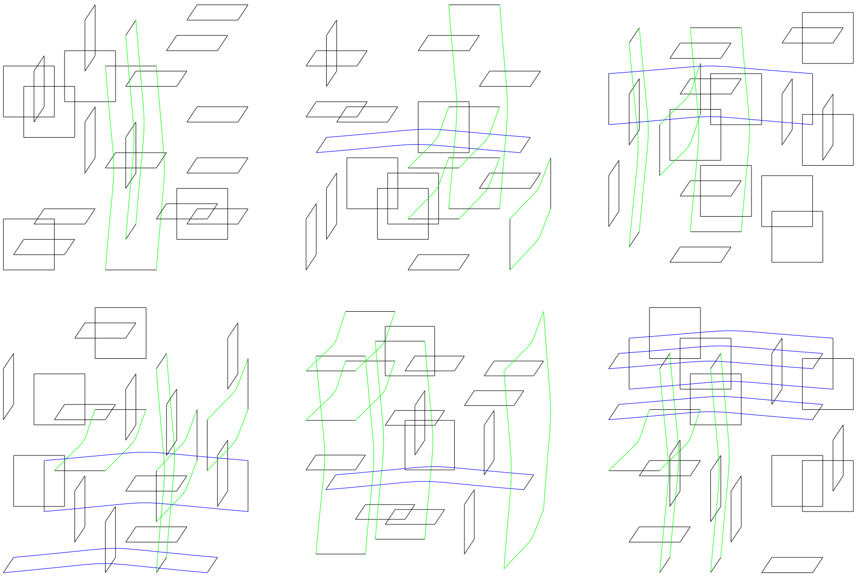
Justification.

1. No collisions, means that bits don't wait until they can read the syndrome.
2. The independence might help when coming to the math.

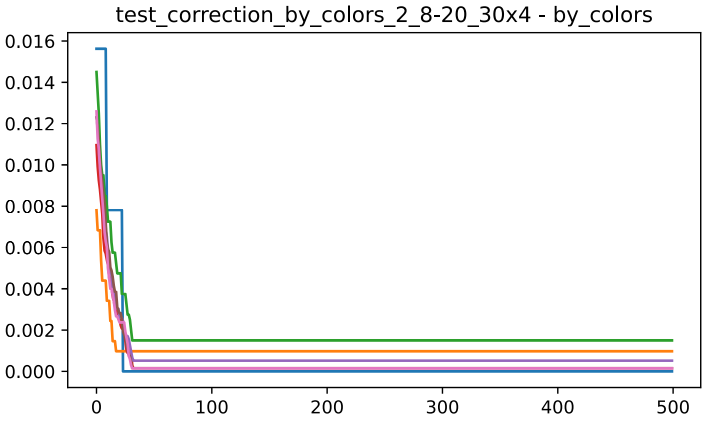
Bottom line.

- I took 32 colors and sample a random coloring.

Colors

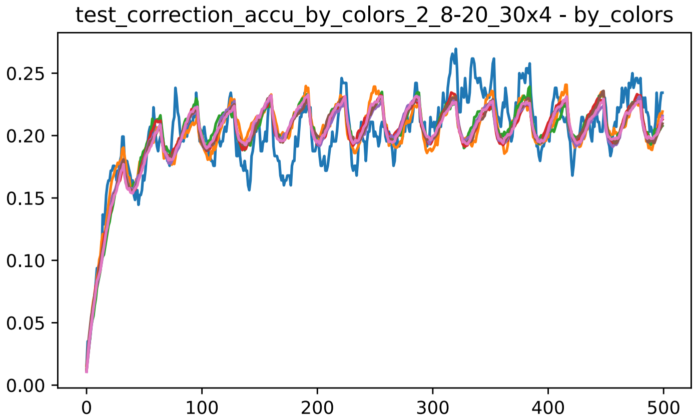


Colors



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Colors



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True

Picking Random Pair

Alg.

Each bit pick two random neighbour checks, and then check if both of them are not satisfied. If so, then flip itself.

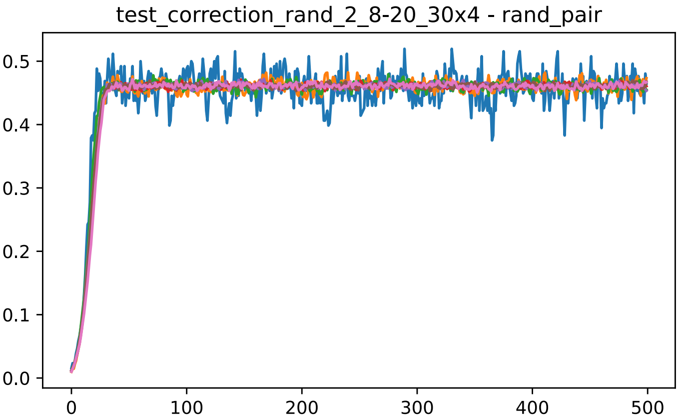
Justification.

With good probability, no too many collisions. Yet, all the bits participate in each round.

Bottom line.

- Fails to correct even in the absence of noise.
- Can we find an analog to $(\frac{1}{2} + \mu)$-majority? Maybe picking 3?

Picking Random



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Swift Rule

Alg.

We define a north vector $\mathbf{1} = (1, 1, 1..1)$, Then each vertex looks only on the cells on its neighbourhood such that they are positive relative to $\mathbf{1}$.
Then, the vertex suggest a correction over those cells, that decreases the local syndrome.

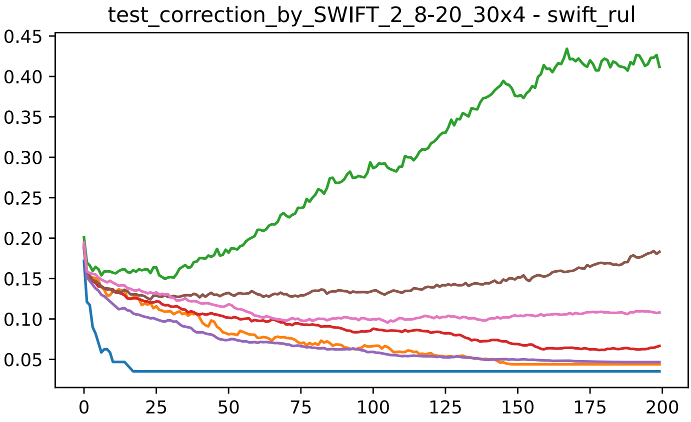
Justification.

1. In the absence of noise, was proven to work.
2. Considered a generalization of Toom's rule.

Bottom line.

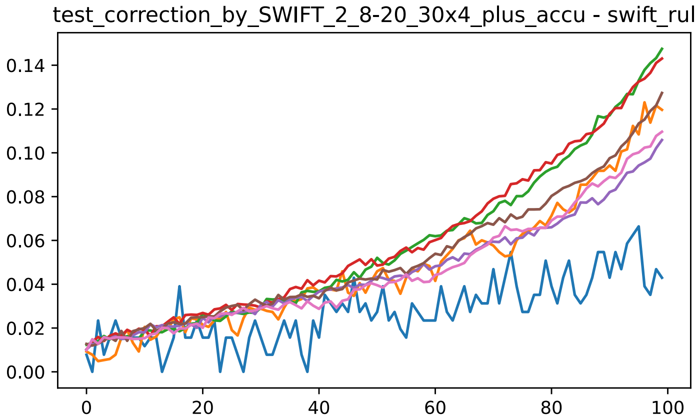
- Works in the absence of noise.
- Simulation are too slow for side length of $n > 16$
- Don't have yet results for accumulating setting.

SWIFT



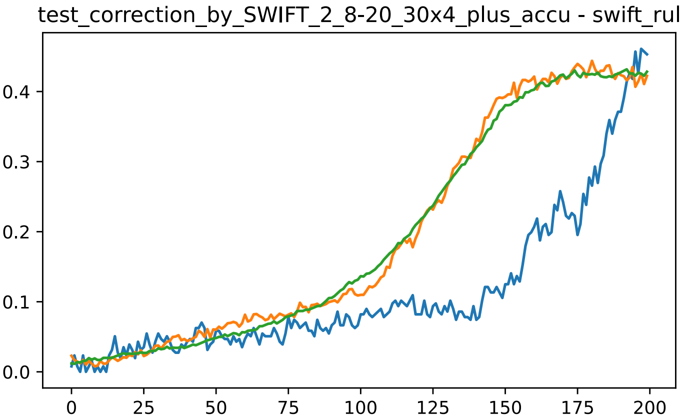
side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.02]
error accumulation :False

SWIFT



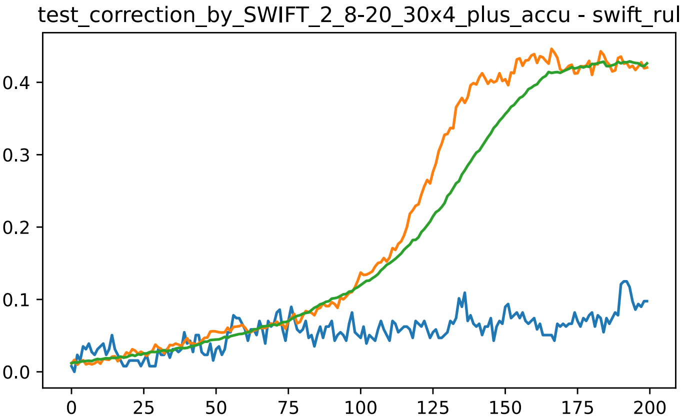
side-length :[8, 16, 30, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True

SWIFT



side-length :[8, 16, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True

SWIFT

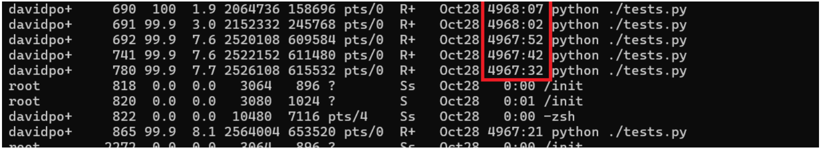


side-length :[8, 16, 40]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True

SWIFT

The Bottleneck

Running a SWIFT simulation requires more than weeks.



```
davidpo+      690  100  1.9 2064736 158696 pts/0   R+   Oct28 4968:07 python ./tests.py
davidpo+      691 99.9  3.0 2152332 245768 pts/0   R+   Oct28 4968:02 python ./tests.py
davidpo+      692 99.9  7.6 2520108 609584 pts/0   R+   Oct28 4967:52 python ./tests.py
davidpo+      741 99.9  7.6 2522152 611480 pts/0   R+   Oct28 4967:42 python ./tests.py
davidpo+      780 99.9  7.7 2526108 615532 pts/0   R+   Oct28 4967:32 python ./tests.py
root          818  0.0  0.0    3064    896 ?       Ss   Oct28    0:00 /init
root          820  0.0  0.0    3080   1024 ?       S    Oct28    0:01 /init
davidpo+      822  0.0  0.0   10480   7116 pts/4   Ss   Oct28    0:00 -zsh
davidpo+      865 99.9  8.1 2564004 653520 pts/0   R+   Oct28 4967:21 python ./tests.py
```

Max Color

Alg.

We change the setting. We back to the setting that allows a non-nosy computation aside, yet we would like to limit the communication to constant number of bits.

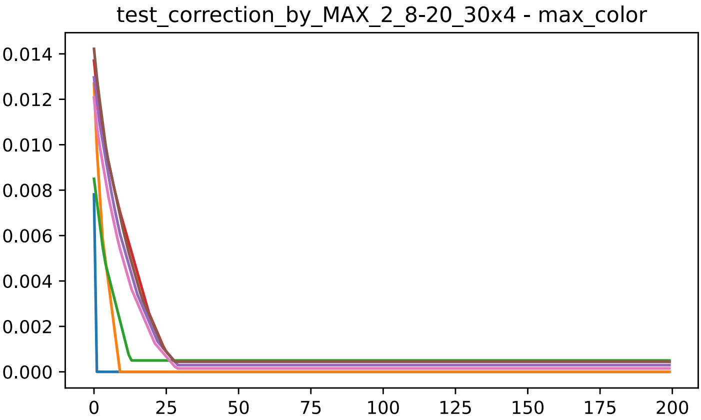
Justification.

1. We don't have works about bounded communication fault tolerance.
2. From mathematical point of view, if the number of colors is c, then we can handle a fraction of $1/c$ of the dirty bits.

Bottom line.

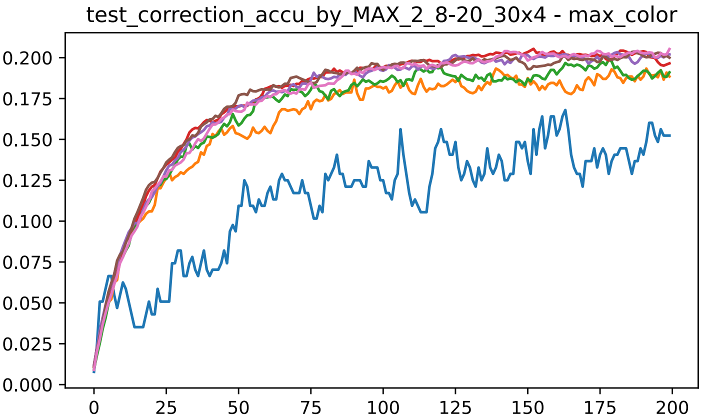
- Works in the absence of noise.
- 'Simulatable'.
- Should be tried when vertices correct according to $(\frac{1}{2} + \mu)$ majority.

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :False

Max Color



side-length :[8, 16, 20, 30, 30, 30, 30]
grid dim :[3, 3, 3, 3, 3, 3, 3]
error rates :[0.001]
error accumulation :True

Ph.D. Future

So What's Next?

1. Moving forward with the experiential work?
   - More experiments, higher statistical standard?
   - Accelerate the decoder? Shifting to $c/c++$? Parallelize it with GPUs/FPGA? Do we have budget? Publishing the source as library?
   - Publishing 'A Tale of Five Decoders'?
2. Clearly, right now, we don't have a memory code. Yet, we do have enough for writing inelegantly on it. Should we write a review paper?
3. What about Bounded-communication Fault Tolerance?
4. Moving to the B stage, can we do it by December?